

100 Kubernetes Troubleshooting Scenarios

Real-World Diagnosis & Fix Guide

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POD ISSUES — Real-World Troubleshooting

#1 Pod Stuck in Pending State

SYMPTOM	Pod remains in Pending indefinitely with no node assignment.
ROOT CAUSE	Insufficient cluster resources (CPU/memory), node selector mismatch, or no nodes available.
FIX	kubectl describe pod <name> — check Events. Scale cluster or adjust resource requests. Verify nodeSelector/tolerations.
<pre>kubectl describe pod <pod-name> -n <namespace></pre>	

#2 CrashLoopBackOff

SYMPTOM	Pod starts, crashes, restarts in a loop with back-off delay.
ROOT CAUSE	Application error, missing env vars/config, misconfigured liveness probe, or OOMKilled.
FIX	Check logs with kubectl logs --previous. Fix app config, increase memory limits, or fix liveness probe thresholds.
<pre>kubectl logs <pod> --previous -n <ns></pre>	

#3 OOMKilled — Pod Memory Limit Exceeded

SYMPTOM	Pod terminates with exit code 137, reason OOMKilled.
ROOT CAUSE	Container exceeded its memory limit set in resources.limits.memory.
FIX	Increase memory limit or optimize application memory usage. Add HPA if load-driven.
<pre>kubectl describe pod <pod> grep -i oom</pre>	

#4 ImagePullBackOff

SYMPTOM	Pod shows ImagePullBackOff or ErrImagePull in status.
ROOT CAUSE	Wrong image name/tag, private registry missing imagePullSecrets, or registry unreachable.
FIX	Verify image exists. Create/attach imagePullSecret. Check network egress to registry.
<pre>kubectl describe pod <pod> grep -A5 Events</pre>	

#5 Pod in Terminating State Forever

SYMPTOM	Pod stays in Terminating for a very long time after deletion.
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ROOT CAUSE	Finalizers not cleared, volume unmount hang, or unresponsive node.
FIX	Force-delete: <code>kubectl delete pod --grace-period=0 --force</code> . Check finalizers and node status.
	<code>kubectl delete pod <pod> --grace-period=0 --force -n <ns></code>

#6 Init Container Failing

SYMPTOM	Pod stuck in Init:0/1 or Init:CrashLoopBackOff.
ROOT CAUSE	Init container script fails — DB not ready, config missing, or wrong command.
FIX	<code>kubectl logs <pod> -c <init-container-name></code> . Fix init logic or add retry/wait loops.
	<code>kubectl logs <pod> -c init-mycontainer -n <ns></code>

#7 Container Exits with Code 1

SYMPTOM	Container terminates immediately with exit code 1.
ROOT CAUSE	Application startup error — missing env variable, bad config file, or runtime exception.
FIX	<code>kubectl logs</code> to see stderr output. Fix application config or missing env vars.
	<code>kubectl logs <pod> -n <ns> --previous</code>

#8 Pod Not Matching Node Affinity

SYMPTOM	Pod stays Pending; describe shows no nodes match affinity rules.
ROOT CAUSE	<code>nodeAffinity requiredDuringSchedulingIgnoredDuringExecution</code> rules don't match any node labels.
FIX	Check node labels: <code>kubectl get nodes --show-labels</code> . Fix affinity spec or label the correct node.
	<code>kubectl get nodes --show-labels grep <label></code>

#9 Pod Evicted Due to Disk Pressure

SYMPTOM	Pod shows Evicted reason with message 'disk pressure'.
ROOT CAUSE	Node disk is full. Kubelet evicts pods when disk usage exceeds eviction threshold.
FIX	Free disk space — clean old images (<code>crictl rmi --prune</code>), remove logs. Add disk or add nodes.
	<code>kubectl describe node <node> grep -i condition</code>

#10 Liveness Probe Failing

SYMPTOM	Pod restarts periodically; describe shows 'Liveness probe failed'.
ROOT CAUSE	App takes too long to respond, probe path wrong, or app is genuinely unhealthy.
FIX	Adjust initialDelaySeconds, periodSeconds, failureThreshold. Verify probe endpoint is correct.
<pre>kubectl describe pod <pod> grep -A10 Liveness</pre>	

#11 Readiness Probe Failing

SYMPTOM	Pod is Running but not receiving traffic; endpoints list is empty.
ROOT CAUSE	Readiness probe returns non-200 or times out — app not yet ready or misconfigured probe.
FIX	Check probe path and port. Increase initialDelaySeconds. Verify app is listening on the probe port.
<pre>kubectl describe pod <pod> grep -A10 Readiness</pre>	

#12 Pod Pending Due to Taint/Toleration Mismatch

SYMPTOM	Describe shows: 0/3 nodes are available: 3 node(s) had taint.
ROOT CAUSE	All nodes have taints that the pod spec does not tolerate.
FIX	Add correct tolerations to pod spec or remove taint from target node.
<pre>kubectl describe node <node> grep Taints</pre>	

#13 Multi-Container Pod — Wrong Container Targeted

SYMPTOM	kubectl logs shows data from wrong container; exec targets wrong process.
ROOT CAUSE	Default container used when pod has multiple containers.
FIX	Always specify -c <container-name> in kubectl logs/exec commands.
<pre>kubectl logs <pod> -c <container-name> -n <ns></pre>	

#14 Pod Running but Application Not Accessible

SYMPTOM	Pod is Running/Ready but requests fail or connection refused.
ROOT CAUSE	App listening on wrong port, service port mismatch, or NetworkPolicy blocking traffic.
FIX	kubectl exec into pod and curl localhost:<port>. Fix containerPort, service targetPort.
<pre>kubectl exec -it <pod> -- curl localhost:8080</pre>	

#15 High Restart Count on Pod

SYMPTOM	Pod shows 15+ restarts in <code>kubectl get pods</code> output.
ROOT CAUSE	Recurring crash — OOM, bad health probes, or intermittent application failure.
FIX	<code>kubectl logs --previous</code> for last crash reason. Profile memory/CPU, fix app stability.
<pre>kubectl get pods -n <ns> --sort-by='.status.containerStatuses[0].restartCount'</pre>	

#16 ConfigMap Not Reflected in Pod

SYMPTOM	Environment variable or mounted file has old value after ConfigMap update.
ROOT CAUSE	Pods cache ConfigMap values at startup; mounted files update eventually, env vars never auto-update.
FIX	Restart pod after ConfigMap change. For dynamic config, use projected volumes with rolling updates.
<pre>kubectl rollout restart deployment/<name> -n <ns></pre>	

#17 Secret Not Mounted in Pod

SYMPTOM	Application reports missing credential or file; volume mount empty.
ROOT CAUSE	Secret name mismatch, namespace mismatch, or secret key name typo in volumeMount.
FIX	<code>kubectl describe pod</code> — check volume mount errors. Verify secret exists in same namespace.
<pre>kubectl get secret <name> -n <ns> -o yaml</pre>	

#18 Pod Cannot Write to Volume

SYMPTOM	Application errors: read-only file system or permission denied on mount path.
ROOT CAUSE	PVC mounted ReadOnlyMany mode, or <code>securityContext.readOnlyRootFilesystem: true</code> .
FIX	Check accessModes on PVC. Set <code>readOnlyRootFilesystem: false</code> or add <code>emptyDir</code> for write paths.
<pre>kubectl describe pvc <name> -n <ns></pre>	

#19 Startup Probe Causing Early Termination

SYMPTOM	Pod killed before app finishes startup; describe shows startupProbe failure.
ROOT CAUSE	<code>failureThreshold * periodSeconds</code> too short for slow-starting applications.
FIX	Increase <code>failureThreshold</code> to allow enough startup time (e.g., $30 * 10s = 5 \text{ min}$).
<pre>kubectl describe pod <pod> grep -A10 Startup</pre>	

#20 Pod Scheduled on Wrong Node

SYMPTOM	Pod lands on a node with wrong hardware/zone, causing performance issues.
ROOT CAUSE	Missing nodeAffinity, nodeSelector, or pod topology spread constraints.
FIX	Add nodeSelector or nodeAffinity rules. Use topologySpreadConstraints for zone distribution.
	<pre>kubectl get pod <pod> -o wide</pre>

DEPLOYMENT ISSUES — Real-World Troubleshooting

#21 Deployment Rollout Stuck

SYMPTOM	kubectl rollout status shows Waiting for deployment rollout to finish.
ROOT CAUSE	New pod fails health checks, insufficient replicas available, or resource quota hit.
FIX	kubectl rollout status -w. Check describe on new pods. Roll back if needed.
	<pre>kubectl rollout undo deployment/<name> -n <ns></pre>

#22 Deployment Not Scaling

SYMPTOM	Replica count in spec is increased but pod count does not change.
ROOT CAUSE	ResourceQuota exceeded, node capacity insufficient, or HPA conflict with manual scale.
FIX	Check quota: kubectl describe quota. Check node capacity. Disable HPA if manual scaling needed.
	<pre>kubectl describe resourcequota -n <ns></pre>

#23 Rolling Update Causes Downtime

SYMPTOM	During update, service has zero available pods briefly.
ROOT CAUSE	maxUnavailable set too high relative to replica count, or readiness probe not configured.
FIX	Set maxUnavailable: 0, maxSurge: 1. Ensure readiness probes are correct so old pods only terminate after new ones are ready.
	<pre>kubectl patch deployment <name> -p '{"spec":{"strategy":{"rollingUpdate":{"maxUnavailable":0}}}}'</pre>

#24 ReplicaSet Pods Not Starting After Image Update

SYMPTOM	New ReplicaSet created but pods fail to start after image change.
ROOT CAUSE	New image has a bug, wrong tag used (e.g., latest pointing to broken image), or registry auth issue.
FIX	Pin image tags. Check new pod logs. Roll back: kubectl rollout undo.
	<pre>kubectl rollout history deployment/<name></pre>

#25 Deployment Shows 0/3 Ready

SYMPTOM	All pods crash immediately; deployment shows 0 ready replicas.
ROOT CAUSE	Fatal application error in new version, bad env config, or missing secret/configmap.
FIX	Check pod events and logs. Verify all referenced Secrets and ConfigMaps exist.
<code>kubectl get events -n <ns> --sort-by='.lastTimestamp'</code>	

#26 HPA Not Scaling Deployment

SYMPTOM	CPU usage high but HPA does not increase replicas.
ROOT CAUSE	Metrics server not installed, metrics.k8s.io API unavailable, or resource requests not set.
FIX	Install metrics-server. Set resource requests on containers. Check HPA describe for conditions.
<code>kubectl describe hpa <name> -n <ns></code>	

#27 Old Pods Not Terminating During Update

SYMPTOM	Both old and new pods running for an extended time during rollout.
ROOT CAUSE	Old pods have long terminationGracePeriodSeconds or preStop hook taking too long.
FIX	Reduce terminationGracePeriodSeconds. Fix preStop hook. Ensure SIGTERM is handled by app.
<code>kubectl describe pod <old-pod> grep TerminationGrace</code>	

#28 Deployment Revision History Full

SYMPTOM	Cannot rollback; error: no revision history found or RevisionHistoryLimit exhausted.
ROOT CAUSE	revisionHistoryLimit set to 0 or very low — old ReplicaSets pruned.
FIX	Set revisionHistoryLimit: 10 in deployment spec to retain rollback history.
<code>kubectl patch deployment <name> -p '{"spec":{"revisionHistoryLimit":10}}'</code>	

#29 Deployment Stuck After Node Drain

SYMPTOM	Pod stays Pending after node drain with PodDisruptionBudget violated.
ROOT CAUSE	PDB minAvailable requires more pods than the cluster can currently schedule.
FIX	Add nodes or temporarily patch PDB. Check: <code>kubectl get pdb -n <ns></code> .
<code>kubectl describe pdb <name> -n <ns></code>	

#30 Environment Variable Not Available in Pod

SYMPTOM	App fails to read env var; echo \$VAR inside pod returns empty.
ROOT CAUSE	Typo in env name in deployment spec, or valueFrom referencing non-existent Secret/ConfigMap key.
FIX	kubectl exec into pod: env grep VAR. Verify deployment env section and referenced resource.
	<pre>kubectl exec <pod> -- env grep MY_VAR</pre>

NETWORKING — Real-World Troubleshooting

#31 Service Not Routing to Pods

SYMPTOM	Requests to ClusterIP service return connection refused or timeout.
ROOT CAUSE	Service selector doesn't match pod labels, wrong targetPort, or pods not Ready.
FIX	kubectl get endpoints <svc>. If empty, fix selector. Verify pod labels match.
	<pre>kubectl get endpoints <svc> -n <ns></pre>

#32 DNS Resolution Failing Inside Pod

SYMPTOM	curl http://my-service returns name resolution failure from inside pod.
ROOT CAUSE	CoreDNS pod not running, NDOTS misconfiguration, or NetworkPolicy blocking DNS (port 53).
FIX	kubectl get pods -n kube-system grep coredns. Check CoreDNS logs. Verify NetworkPolicy allows DNS.
	<pre>kubectl exec <pod> -- nslookup my-service.default.svc.cluster.local</pre>

#33 NodePort Service Not Accessible Externally

SYMPTOM	Hitting NodeIP:NodePort from outside cluster returns connection refused.
ROOT CAUSE	Security group / firewall blocking the NodePort range (30000-32767), or service not correctly configured.
FIX	Add firewall rule for NodePort. Verify service type: NodePort. Check node IP is reachable.
	<pre>kubectl get svc <name> -o wide</pre>

#34 LoadBalancer Service Stuck in Pending

SYMPTOM	Service of type LoadBalancer shows <pending> for EXTERNAL-IP indefinitely.
ROOT CAUSE	Cloud controller manager not running, insufficient cloud quota, or missing IAM permissions for LB creation.
FIX	Check CCM pods in kube-system. Verify cloud provider IAM role. Check cloud console for LB creation errors.
	<pre>kubectl describe svc <name> -n <ns> grep -A10 Events</pre>

#35 Ingress Not Routing Traffic

SYMPTOM	Ingress resource exists but requests return 404 or default backend.
ROOT CAUSE	Ingress class annotation missing/wrong, host mismatch, or ingress controller not deployed.
FIX	Verify ingressClassName. Confirm ingress controller pods are running. Check host field matches request.
	<code>kubectl describe ingress <name> -n <ns></code>

#36 TLS Certificate Not Served by Ingress

SYMPTOM	Browser shows certificate error; HTTPS returns wrong cert or HTTP fallback.
ROOT CAUSE	TLS secret missing, cert-manager not issuing certificate, or wrong secretName in ingress spec.
FIX	<code>kubectl describe certificate -n <ns></code> . Check cert-manager logs. Ensure ClusterIssuer/Issuer exists.
	<code>kubectl get certificate -n <ns></code>

#37 Pod-to-Pod Communication Failing

SYMPTOM	Pod A cannot curl or ping Pod B on its pod IP.
ROOT CAUSE	CNI plugin misconfigured, NetworkPolicy blocking pod-to-pod traffic, or overlay network issue.
FIX	Check CNI pods (calico/flannel/cilium). <code>kubectl describe networkpolicy</code> . Test with netshoot debug pod.
	<code>kubectl run netshoot --rm -it --image nicolaka/netshoot -- bash</code>

#38 NetworkPolicy Accidentally Blocking All Traffic

SYMPTOM	After applying NetworkPolicy, all pods lose connectivity.
ROOT CAUSE	Default-deny policy applied without proper allow rules for required traffic.
FIX	<code>kubectl describe networkpolicy</code> . Add explicit allow rules for required ingress/egress. Check DNS port 53 is allowed.
	<code>kubectl get networkpolicy -n <ns></code>

#39 Intermittent Service Timeouts

SYMPTOM	Service calls succeed most of the time but randomly timeout.
ROOT CAUSE	A subset of backend pods are unhealthy, kube-proxy iptables rules stale, or connection tracking table full.
FIX	Check readiness probes on all pods. Restart kube-proxy. Inspect conntrack table on node.

```
kubectl get pods -n <ns> -o wide | grep -v Running
```

#40 ExternalName Service Not Resolving

SYMPTOM	Service of type ExternalName returns NXDOMAIN or wrong IP.
ROOT CAUSE	CNAME chain too long, external DNS flapping, or CoreDNS not forwarding external queries.
FIX	kubectl exec into pod and nslookup the external name. Check CoreDNS ConfigMap forward rules.
<pre>kubectl get configmap coredns -n kube-system -o yaml</pre>	

#41 Headless Service Returning Wrong IPs

SYMPTOM	DNS for headless service returns stale pod IPs after pod restarts.
ROOT CAUSE	DNS TTL caching in application or pod, or endpoints controller lag.
FIX	Reduce DNS TTL in CoreDNS or application DNS cache. Verify endpoint update latency.
<pre>kubectl get endpoints <svc> -n <ns> -w</pre>	

#42 Service Account Token Volume Projection Error

SYMPTOM	Pod fails to start: token volume projection error in events.
ROOT CAUSE	TokenRequest API disabled, or projected service account token expiry misconfigured.
FIX	Ensure TokenRequest feature gate is enabled (default in 1.20+). Check RBAC for system:serviceaccounts.
<pre>kubectl describe pod <pod> grep -i token</pre>	

#43 Cross-Namespace Service Access Failing

SYMPTOM	Pod in namespace A cannot reach service in namespace B.
ROOT CAUSE	NetworkPolicy in namespace B blocking ingress from other namespaces, or FQDN not used.
FIX	Use FQDN: <svc>.<namespace>.svc.cluster.local. Add NetworkPolicy ingress rule from namespace A.
<pre>kubectl exec <pod-ns-a> -- curl http://svc-b.namespace-b.svc.cluster.local</pre>	

#44 Kube-Proxy Not Updating iptables Rules

SYMPTOM	New service endpoints not reachable; old pod IPs still in use.
ROOT CAUSE	kube-proxy pod crashed or stuck, or iptables sync interval too high.
FIX	Check kube-proxy logs: <code>kubectl logs -n kube-system</code> . Restart kube-proxy DaemonSet.
<pre>kubectl rollout restart daemonset/kube-proxy -n kube-system</pre>	

#45 Egress Traffic Blocked from Pod

SYMPTOM	Pod cannot reach external internet or external APIs.
ROOT CAUSE	NetworkPolicy with egress rules missing, or cloud security group/NAT gateway misconfigured.
FIX	Add egress allow rule in NetworkPolicy for external CIDR. Verify NAT gateway or IGW in cloud VPC.
<pre>kubectl exec <pod> -- curl -v https://google.com</pre>	

STORAGE — Real-World Troubleshooting

#46 PVC Stuck in Pending

SYMPTOM	PersistentVolumeClaim stays Pending; no PV bound.
ROOT CAUSE	No StorageClass available, provisioner not running, or access mode mismatch.
FIX	kubectl describe pvc. Check StorageClass exists. Verify provisioner pod is running.
	<pre>kubectl get storageclass && kubectl describe pvc <name> -n <ns></pre>

#47 PV Not Releasing After PVC Deletion

SYMPTOM	PersistentVolume stuck in Released state; cannot be rebound.
ROOT CAUSE	PV reclaim policy is Retain — manual intervention required to rebind.
FIX	Edit PV: remove claimRef to make it Available, or delete and re-create PV.
	<pre>kubectl patch pv <pv-name> -p '{"spec":{"claimRef":null}}'</pre>

#48 Volume Mount Permission Denied

SYMPTOM	Application gets permission denied accessing mounted volume path.
ROOT CAUSE	fsGroup/runAsUser not set; mounted volume owned by root, app runs as non-root user.
FIX	Set securityContext.fsGroup to GID that owns the files, or use initContainer to chmod the mount.
	<pre>kubectl exec <pod> -- ls -la /data</pre>

#49 EBS Volume Stuck Attaching on AWS EKS

SYMPTOM	Pod pending with AttachVolume.Attach failed; volume stuck attaching.
ROOT CAUSE	Volume still attached to previous node (not detached cleanly), or AZ mismatch between pod and EBS volume.
FIX	Manually detach EBS in AWS console. Ensure PV and node are in same AZ using topologyKey.
	<pre>kubectl describe pod <pod> grep -A5 AttachVolume</pre>

#50 StatefulSet Pod Can't Find PVC

SYMPTOM	StatefulSet pod starts but crashes with volume not found error.
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ROOT CAUSE	VolumeClaimTemplate name changed, PVC manually deleted, or namespace mismatch.
FIX	Restore PVC with exact naming convention: <template-name>--<statefulset-name>--<ordinal>.
	<code>kubectl get pvc -n <ns> grep <statefulset-name></code>

#51 CSI Driver Not Installed

SYMPTOM	PVC Pending with error: no plugin found for CSI driver.
ROOT CAUSE	CSI driver (EBS CSI, EFS CSI, etc.) not installed on cluster.
FIX	Install the CSI driver via Helm or manifests. Verify DaemonSet pods are running.
	<code>kubectl get pods -n kube-system grep csi</code>

#52 Disk Full on Node Causing Pod Eviction

SYMPTOM	Multiple pods evicted; node shows DiskPressure condition.
ROOT CAUSE	Logs, images, or application writes filling node disk.
FIX	Clean images: <code>crictl rmi --prune</code> . Set log rotation. Add log size limits to container spec.
	<code>crictl rmi --prune && df -h</code>

#53 ReadWriteMany PVC Not Supported

SYMPTOM	Deployment with multiple replicas fails to start all pods; PVC can only bind once.
ROOT CAUSE	StorageClass (e.g., EBS gp2) only supports ReadWriteOnce access mode.
FIX	Use NFS or EFS-based StorageClass that supports ReadWriteMany. Or use one pod writing to shared storage.
	<code>kubectl get storageclass -o yaml grep allowVolumeExpansion</code>

#54 Volume Expansion Not Taking Effect

SYMPTOM	PVC size patched to larger value but filesystem inside pod still shows old size.
ROOT CAUSE	Volume expanded at PV level but filesystem inside pod not resized.
FIX	Restart pod to trigger online resize, or mount with <code>allowVolumeExpansion: true</code> in StorageClass.
	<code>kubectl patch pvc <name> -p '{"spec":{"resources":{"requests":{"storage":"20Gi"}}}}'</code>

#55 emptyDir Volume Data Lost After Pod Restart

SYMPTOM	Application data lost after pod restart; files written to emptyDir are gone.
ROOT CAUSE	emptyDir is ephemeral and tied to pod lifecycle — data is wiped on pod termination.
FIX	Use PVC for persistent data. emptyDir is only for scratch space or inter-container sharing.
	<code>kubectl describe pod <pod> grep -A5 Volumes</code>

RBAC & SECURITY — Real-World Troubleshooting

#56 Forbidden Error When Accessing Resources

SYMPTOM	kubectl or application gets Error: pods is forbidden: User cannot list resource.
ROOT CAUSE	ServiceAccount or user missing Role/ClusterRole binding for the required verb.
FIX	Create RoleBinding/ClusterRoleBinding. Use kubectl auth can-i to verify permissions.
	<pre>kubectl auth can-i list pods --as=system:serviceaccount:<ns>:<sa></pre>

#57 ServiceAccount Token Not Automounted

SYMPTOM	Application inside pod cannot call Kubernetes API — token file missing at /var/run/secrets/.
ROOT CAUSE	automountServiceAccountToken: false set on ServiceAccount or pod spec.
FIX	Set automountServiceAccountToken: true on pod or ServiceAccount.
	<pre>kubectl get serviceaccount <sa> -n <ns> -o yaml</pre>

#58 Pod Security Admission Blocking Pod Creation

SYMPTOM	Pod creation fails: pod violates PodSecurity policy enforce level.
ROOT CAUSE	Pod spec uses privileged mode or hostPath mounts that violate namespace PSA policy.
FIX	Remove privileged/hostPath if not needed. Or update namespace label to appropriate PSA level.
	<pre>kubectl get namespace <ns> -o yaml grep pod-security</pre>

#59 RBAC Roles Not Propagated Across Namespaces

SYMPTOM	User has ClusterRole but cannot access resources in specific namespace.
ROOT CAUSE	ClusterRole bound with RoleBinding (namespace-scoped) instead of ClusterRoleBinding.
FIX	Use ClusterRoleBinding for cluster-wide access. Use RoleBinding per-namespace for scoped access.
	<pre>kubectl get rolebindings,clusterrolebindings --all-namespaces grep <user></pre>

#60 Audit Logs Showing Unauthorized API Calls

SYMPTOM	Audit log shows 403s for service account trying to list/create resources.
ROOT CAUSE	Newly deployed application making API calls without proper RBAC setup.
FIX	Identify SA from audit log. Create Role + RoleBinding for required verbs. Apply least privilege.
	<pre>kubectl logs -n kube-system <apiserver-pod> grep audit</pre>

NODE ISSUES — Real-World Troubleshooting

#61 Node in NotReady State

SYMPTOM	kubectl get nodes shows node as NotReady; pods on it not scheduling.
ROOT CAUSE	kubelet crashed/stopped, node network issue, or node is fully resource-exhausted.
FIX	SSH to node: systemctl status kubelet. Restart kubelet. Check /var/log/kubelet.log.
	<pre>systemctl restart kubelet && journalctl -u kubelet -n 50</pre>

#62 Node Under Memory Pressure

SYMPTOM	Node shows MemoryPressure=True condition; pods being evicted.
ROOT CAUSE	Total memory consumed by pods exceeds eviction threshold.
FIX	Identify top consumers: kubectl top pods -n <ns>. Set memory limits. Add nodes.
	<pre>kubectl top pods --all-namespaces --sort-by=memory head -20</pre>

#63 Node Under CPU Pressure

SYMPTOM	Applications throttled; node CPU usage at 100%.
ROOT CAUSE	No CPU limits set; one workload consuming all CPU, throttling others.
FIX	Set CPU limits and requests. Use LimitRange to enforce defaults. Scale horizontally.
	<pre>kubectl top nodes && kubectl top pods --all-namespaces --sort-by=cpu head -20</pre>

#64 DaemonSet Pod Not Scheduled on New Node

SYMPTOM	New node added to cluster but DaemonSet pod not appearing on it.
ROOT CAUSE	New node has taint that DaemonSet doesn't tolerate, or nodeSelector doesn't match.
FIX	Add toleration for node taint in DaemonSet spec. Verify DaemonSet nodeSelector.
	<pre>kubectl describe daemonset <name> -n <ns> grep -i toleration</pre>

#65 Node Cordoned — Pods Not Scheduling

SYMPTOM	Pods stay Pending; describe shows node is cordoned (SchedulingDisabled).
ROOT CAUSE	Node was manually cordoned with kubectl cordon for maintenance and not uncordoned.

FIX	kubectl uncordon <node-name> to re-enable scheduling.
	<code>kubectl uncordon <node-name></code>

#66 Node Disk I/O Saturation

SYMPTOM	Node I/O wait high; pods experiencing slow disk reads/writes.
ROOT CAUSE	Multiple I/O-heavy workloads competing for disk, or underlying volume throughput limit hit.
FIX	Move I/O-intensive workloads to nodes with faster storage. Set I/O limits with cgroup blkio.
	<code>iostat -xz 1 && kubectl top nodes</code>

#67 Kubelet Certificate Expired

SYMPTOM	Node shows NotReady; kubelet log shows certificate expired or x509 error.
ROOT CAUSE	TLS certificate used by kubelet expired and auto-renewal failed.
FIX	Renew certificates: kubeadm certs renew all. Restart kubelet and control plane components.
	<code>kubeadm certs check-expiration</code>

#68 Node Clock Skew Causing API Errors

SYMPTOM	API server returns 401 errors; JWT validation fails; etcd clock skew warnings.
ROOT CAUSE	Node system clock drifted more than 5 minutes from API server time.
FIX	Sync NTP: timedatectl status. Configure chronyd or ntpd on all nodes.
	<code>timedatectl status && chronyc tracking</code>

CONTROL PLANE — Real-World Troubleshooting

#69 API Server Not Responding

SYMPTOM	kubectl commands hang or return connection refused.
ROOT CAUSE	kube-apiserver pod crashed, etcd unreachable, or control plane node overloaded.
FIX	On control plane: check /var/log/kube-apiserver.log. Restart static pod. Check etcd health.
<pre>kubectl get --raw='/healthz' && crictl ps grep apiserver</pre>	

#70 etcd High Latency / Slow Writes

SYMPTOM	API server slow; etcd logs show slow heartbeats and leader election warnings.
ROOT CAUSE	Slow disk on etcd nodes, large etcd database, or too many watchers.
FIX	Use SSD for etcd data dir. Compact/defrag etcd. Reduce number of CRD instances.
<pre>etcdctl endpoint status --write-out=table -- endpoints=https://127.0.0.1:2379</pre>	

#71 Scheduler Not Placing Pods

SYMPTOM	Pods stay Pending even with available nodes; no scheduling events in describe.
ROOT CAUSE	kube-scheduler crashed, misconfigured scheduling profile, or scheduler plugin conflict.
FIX	Check scheduler pod: kubectl get pods -n kube-system grep scheduler. Review logs.
<pre>kubectl logs -n kube-system <kube-scheduler-pod></pre>	

#72 Controller Manager Not Reconciling

SYMPTOM	Deployments not creating pods; ReplicaSets not scaling; Jobs not starting.
ROOT CAUSE	kube-controller-manager is down or stuck in leader election.
FIX	kubectl get pods -n kube-system grep controller-manager. Check logs for panic/errors.
<pre>kubectl logs -n kube-system kube-controller-manager-<node></pre>	

#73 etcd Member Out of Quorum

SYMPTOM	etcd cluster has no leader; API server returns etcd cluster is unavailable.
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ROOT CAUSE	One or more etcd members crashed, reducing quorum below majority (need $>n/2$ healthy).
FIX	Restore crashed etcd member from snapshot. Or remove failed member and re-add.
<pre>etcdctl member list --endpoints=https://127.0.0.1:2379</pre>	

#74 Certificate Signing Request Stuck Pending

SYMPTOM	New node cannot join; CSR shows Pending in <code>kubectl get csr</code> .
ROOT CAUSE	kube-controller-manager not approving CSRs (auto-approval disabled), or RBAC issue.
FIX	Manually approve: <code>kubectl certificate approve <csr-name></code> . Or enable controller-based auto-approval.
<pre>kubectl get csr && kubectl certificate approve <name></pre>	

#75 Admission Webhook Blocking Pod Creation

SYMPTOM	Pod creation fails with admission webhook rejected or webhook timeout error.
ROOT CAUSE	Custom admission webhook returning reject, or webhook server unreachable causing timeout.
FIX	<code>kubectl get validatingwebhookconfigurations</code> . Check webhook service health. Set failurePolicy: Ignore temporarily.
<pre>kubectl get validatingwebhookconfigurations -o yaml</pre>	

HELM & CONFIG — Real-World Troubleshooting

#76 Helm Release Stuck in Pending-Install

SYMPTOM	helm status shows PENDING-INSTALL; resources deployed but release not finalized.
ROOT CAUSE	Helm hook job failed, or previous failed install left partial state.
FIX	helm rollback or helm uninstall --no-hooks. Delete stuck hook jobs manually.
	<pre>helm history <release> -n <ns></pre>

#77 Helm Values Not Applied to Deployment

SYMPTOM	Helm upgrade succeeds but pod environment has old values.
ROOT CAUSE	Cached deployment not rolling — pods not restarted after config-only helm upgrade.
FIX	Use helm upgrade --force or add checksum annotation to deployment to trigger rollout on config change.
	<pre>helm upgrade <release> . --set image.tag=new --force</pre>

#78 Namespace Not Created Before Helm Install

SYMPTOM	helm install fails: namespace <ns> not found.
ROOT CAUSE	--create-namespace flag not used and namespace does not exist.
FIX	kubectl create namespace <ns> first, or use helm install --create-namespace.
	<pre>helm install <name> <chart> -n <ns> --create-namespace</pre>

#79 ConfigMap Key Not Found Error

SYMPTOM	Pod fails with error: key not found in ConfigMap; envFrom producing empty env.
ROOT CAUSE	ConfigMap key name typo in valueFrom.configMapKeyRef, or ConfigMap was re-created with different keys.
FIX	kubectl describe configmap <name>. Verify exact key names. Update deployment env section.
	<pre>kubectl get configmap <name> -n <ns> -o jsonpath='{.data}'</pre>

#80 Duplicate Resource Conflict After Helm Upgrade

SYMPTOM	Helm upgrade fails: resource already exists or cannot patch immutable field.
ROOT CAUSE	Resource was manually modified or created outside Helm, causing ownership conflict.

FIX	kubectl annotate or label the resource with Helm ownership, or use --force flag.
	<pre>helm upgrade <release> <chart> --force -n <ns></pre>

CI/CD & GITOPS — Real-World Troubleshooting

#81 ArgoCD App Out of Sync After Deployment

SYMPTOM	ArgoCD shows application OutOfSync; manual changes in cluster not matching Git.
ROOT CAUSE	Manual kubectl edit or patch bypassed GitOps workflow, causing drift.
FIX	argocd app sync <app> --force to reset to Git state. Enforce GitOps — disable direct kubectl edits.
<pre>argocd app sync <app-name> --force</pre>	

#82 ArgoCD Stuck in Progressing State

SYMPTOM	ArgoCD shows Progressing for a long time; deployment never completes.
ROOT CAUSE	Pod health check failing, deployment rollout stuck, or ArgoCD health check misconfigured.
FIX	Check deployment status in ArgoCD UI. Check pod events. Fix deployment issue.
<pre>argocd app get <app> --show-operation</pre>	

#83 FluxCD Kustomization Fails to Apply

SYMPTOM	Flux shows reconciliation failed; resources not created in cluster.
ROOT CAUSE	YAML syntax error in kustomization, missing dependency, or RBAC insufficient for Flux SA.
FIX	flux get kustomizations. Check flux logs: kubectl logs -n flux-system deploy/kustomize-controller.
<pre>flux get kustomizations --all-namespaces</pre>	

#84 Pipeline Fails on Docker Build Step

SYMPTOM	CI pipeline fails at docker build: permission denied to push or image pull rate limit.
ROOT CAUSE	Registry auth token expired, missing imagePullSecret, or Docker Hub rate limit.
FIX	Refresh registry credentials. Use private registry. Add imagePullSecret to pipeline service account.
<pre>kubectl create secret docker-registry regcred --docker-server=... -n <ns></pre>	

#85 Canary Deployment Not Splitting Traffic

SYMPTOM	All traffic goes to stable version; canary pods exist but receive no requests.
ROOT CAUSE	Service mesh (Istio/Linkerd) VirtualService not configured, or Ingress weights not set.
FIX	Configure Istio VirtualService with weight: 90/10. Or use Argo Rollouts canary strategy.
<pre>kubectl get virtualservice -n <ns> -o yaml</pre>	

OBSERVABILITY — Real-World Troubleshooting

#86 Prometheus Not Scraping Pod Metrics

SYMPTOM	Pod metrics missing in Prometheus; targets show DOWN.
ROOT CAUSE	Missing prometheus.io/scrape: 'true' annotation, wrong port annotation, or NetworkPolicy blocking scrape.
FIX	Add annotations to pod/service. Verify metricsPort. Allow Prometheus namespace in NetworkPolicy.
<pre>kubectl annotate pod <pod> prometheus.io/scrape=true prometheus.io/port=8080</pre>	

#87 kubectl top Showing Unknown Metrics

SYMPTOM	kubectl top nodes/pods shows unknown or no metrics.
ROOT CAUSE	Metrics Server not installed or pod not running.
FIX	kubectl apply -f metrics-server.yaml or install via Helm. Verify metrics-server pod is Running.
<pre>kubectl get pods -n kube-system grep metrics-server</pre>	

#88 Grafana Dashboard Showing No Data

SYMPTOM	Grafana panels display No Data despite Prometheus running.
ROOT CAUSE	Datasource URL wrong, time range mismatch, or PromQL query incorrect.
FIX	Test datasource in Grafana. Check time range. Validate PromQL with Prometheus UI directly.
<pre>kubectl port-forward svc/prometheus 9090:9090 -n monitoring</pre>	

#89 Log Aggregator Missing Container Logs

SYMPTOM	Fluentd/Fluentbit not shipping all pod logs to Elasticsearch/Loki.
ROOT CAUSE	Log aggregator DaemonSet not on all nodes, log path mismatch, or buffer overflow.
FIX	Check DaemonSet coverage: kubectl get pods -n logging -o wide. Fix log path. Increase buffer size.
<pre>kubectl logs -n logging <fluentbit-pod> grep -i error</pre>	

#90 Alert Manager Not Sending Notifications

SYMPTOM	Prometheus alert fires but no notification received in Slack/PagerDuty.
ROOT CAUSE	AlertManager config YAML error, webhook URL wrong, or inhibit rule suppressing alert.
FIX	kubectl get secret alertmanager-config -n monitoring. Validate YAML. Check AlertManager UI silence/inhibit.
	<pre>kubectl port-forward svc/alertmanager 9093:9093 -n monitoring</pre>

ADVANCED — Real-World Troubleshooting**#91 CRD Version Mismatch After Upgrade**

SYMPTOM	Operator reconciliation fails; error: no kind CustomResource is registered.
ROOT CAUSE	CRD schema version not upgraded after operator upgrade; stored versions mismatch.
FIX	Re-apply updated CRD manifests. Run storage migration if stored version deprecated.
<pre>kubectl get crd <name> -o jsonpath='{.status.storedVersions}'</pre>	

#92 Cluster Autoscaler Not Removing Idle Nodes

SYMPTOM	Empty nodes with no pods still running; cluster autoscaler not scaling down.
ROOT CAUSE	System pods or DaemonSets prevent scale-down, or scale-down-unneeded-time not elapsed.
FIX	Check CA logs. Annotate non-scalable pods. Verify cluster-autoscaler parameters.
<pre>kubectl logs -n kube-system -l app=cluster-autoscaler grep scale-down</pre>	

#93 KEDA ScaledObject Not Triggering Scale

SYMPTOM	Workload not scaling despite metric threshold crossed.
ROOT CAUSE	KEDA operator not running, trigger auth missing, or metric source unavailable.
FIX	<code>kubectl describe scaledobject <name></code> . Check KEDA operator pod logs. Verify TriggerAuthentication.
<pre>kubectl get scaledobject -n <ns> && kubectl describe scaledobject <name></pre>	

#94 Istio Sidecar Injection Not Working

SYMPTOM	Pods not getting Envoy sidecar; mesh traffic control not applied.
ROOT CAUSE	Namespace label istio-injection: enabled missing, or pod annotation sidecar.istio.io/inject: false set.
FIX	<code>kubectl label namespace <ns> istio-injection=enabled</code> . Re-create pods.
<pre>kubectl label namespace <ns> istio-injection=enabled</pre>	

#95 Velero Backup Failing

SYMPTOM	Velero backup job shows Failed status; PVC snapshots missing.
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ROOT CAUSE	Cloud snapshot permissions missing, Velero plugin not installed, or backup storage location unavailable.
FIX	Check velero describe backup. Verify IAM permissions for volume snapshot. Check BSL status.
<pre>velero backup describe <backup-name> --details</pre>	

#96 PodDisruptionBudget Blocking All Node Drains

SYMPTOM	kubectl drain <node> hangs; cannot evict pod as it would violate PDB.
ROOT CAUSE	PDB minAvailable equals total replicas; no replica can be evicted.
FIX	Temporarily scale up deployment before drain. Or patch PDB. Plan PDB to allow at least 1 eviction.
<pre>kubectl get pdb -n <ns> && kubectl drain <node> --ignore-daemonsets</pre>	

#97 Vertical Pod Autoscaler Causing Restart Loop

SYMPTOM	VPA keeps restarting pods to apply new resource recommendations.
ROOT CAUSE	VPA in Auto mode continuously updating limits; app takes time to stabilize.
FIX	Use VPA in Off or Initial mode first to observe recommendations without forced restarts.
<pre>kubectl describe vpa <name> -n <ns></pre>	

#98 Namespace Stuck in Terminating

SYMPTOM	kubectl delete namespace <ns> runs but namespace stays Terminating forever.
ROOT CAUSE	Finalizers on resources inside namespace not cleared, or API service unavailable.
FIX	Find stuck resources: kubectl api-resources --verbs=list -o name xargs -l{} kubectl get {} -n <ns>. Patch finalizers.
<pre>kubectl get namespace <ns> -o json jq '.spec.finalizers=[]' kubectl replace --raw /api/v1/namespaces/<ns>/finalize -f -</pre>	

#99 Resource Quota Blocking New Deployments

SYMPTOM	Pod creation denied: exceeded quota: requests.cpu, must be less than or equal to.
ROOT CAUSE	Namespace ResourceQuota exhausted; no more CPU/memory can be allocated.
FIX	Increase quota or reduce existing resource requests. Clean up unused workloads.
<pre>kubectl describe resourcequota -n <ns></pre>	

#100 Job Pods Not Cleaning Up After Completion

SYMPTOM	Completed job pods accumulating; cluster full of Completed pods.
ROOT CAUSE	ttlSecondsAfterFinished not set on Job spec.
FIX	Set ttlSecondsAfterFinished: 300 on Job spec. For existing jobs, delete manually.
<pre>kubectrl delete pods --field-selector=status.phase==Succeeded -n <ns></pre>	